Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

Digital Assignment Lab - 4

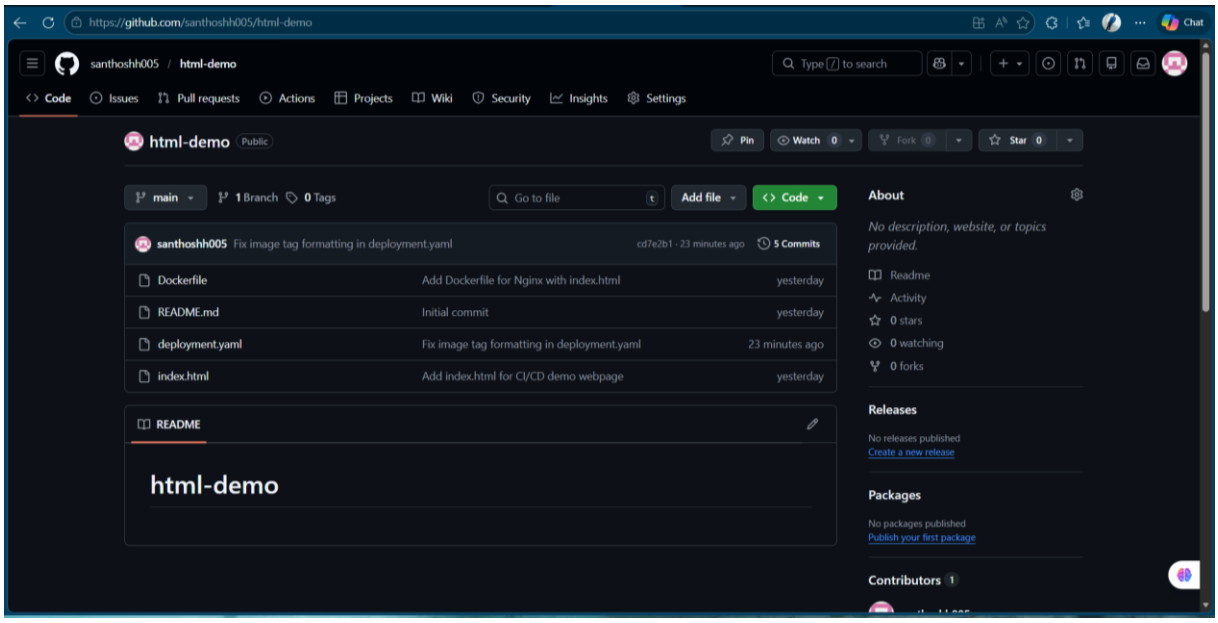
Course Name :- Agile Development Process and Devops Lab

Course code :- ISWE406P

Name :- G SAI SANTHOSH

Reg-No:- 23MIS0227

1) Create a repository in GIT and create the following files:
   - index.html
   - Dockerfile
   - Deployment.yaml



The code for the three files are given below:

Create index.html:

```
<!DOCTYPE html>
<html>
<head>
<title>CI/CD Demo</title>
<style>
body{
text-align:center;
font-family:Times New Roman;
background-color:#af0d0d;
}
```

```
h1{
color:#0ef777;
}
</style>
</head>
<body>
<h1>CI/CD Pipeline Working</h1>
<h2>Deployed using Jenkins + Docker + Kubernetes</h2>
<p>This webpage is automatically deployed using a CI/CD pipeline.</p>
</body>
</html>
```

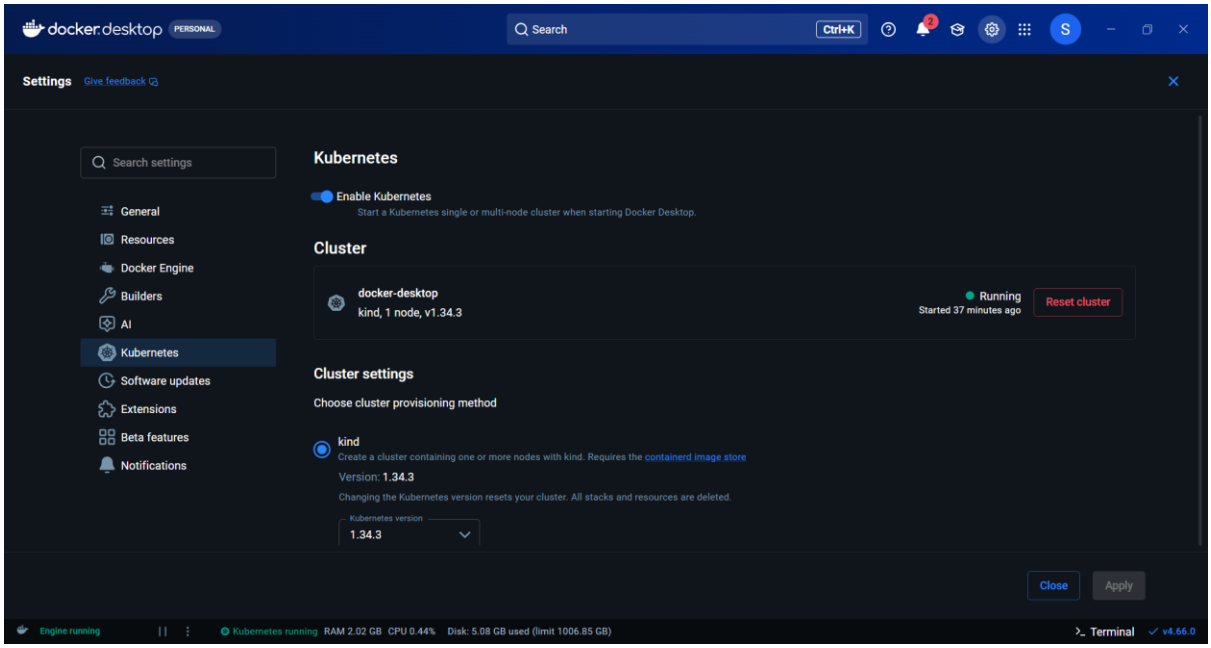
Create a Dockerfile

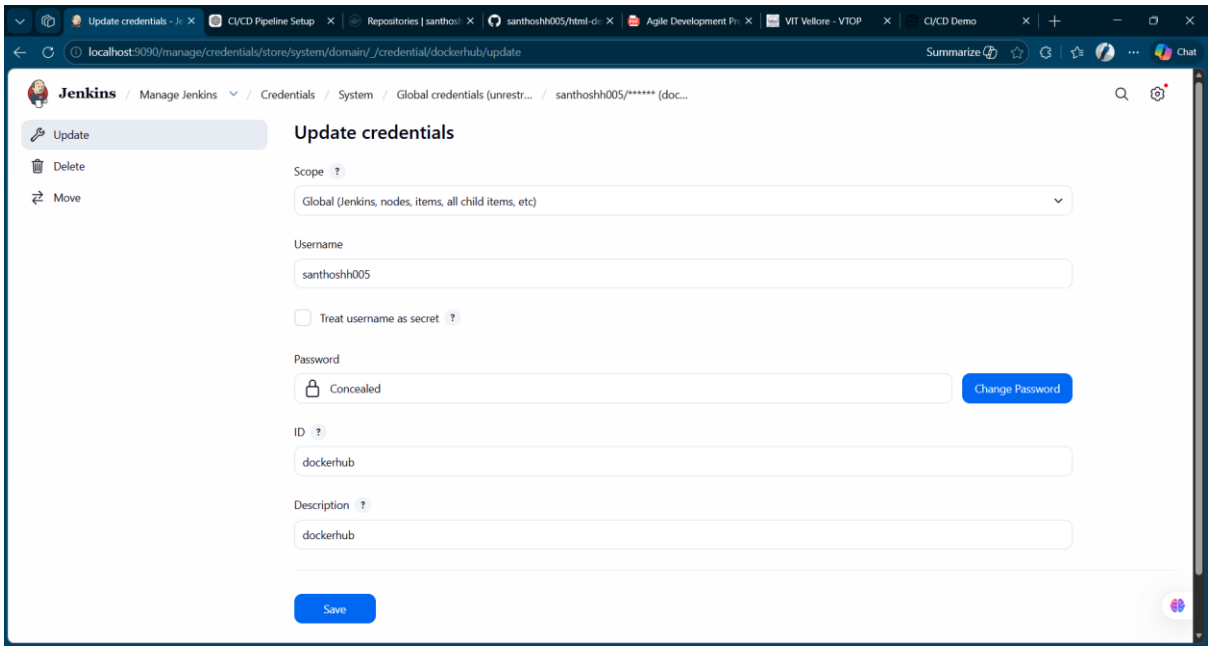
```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

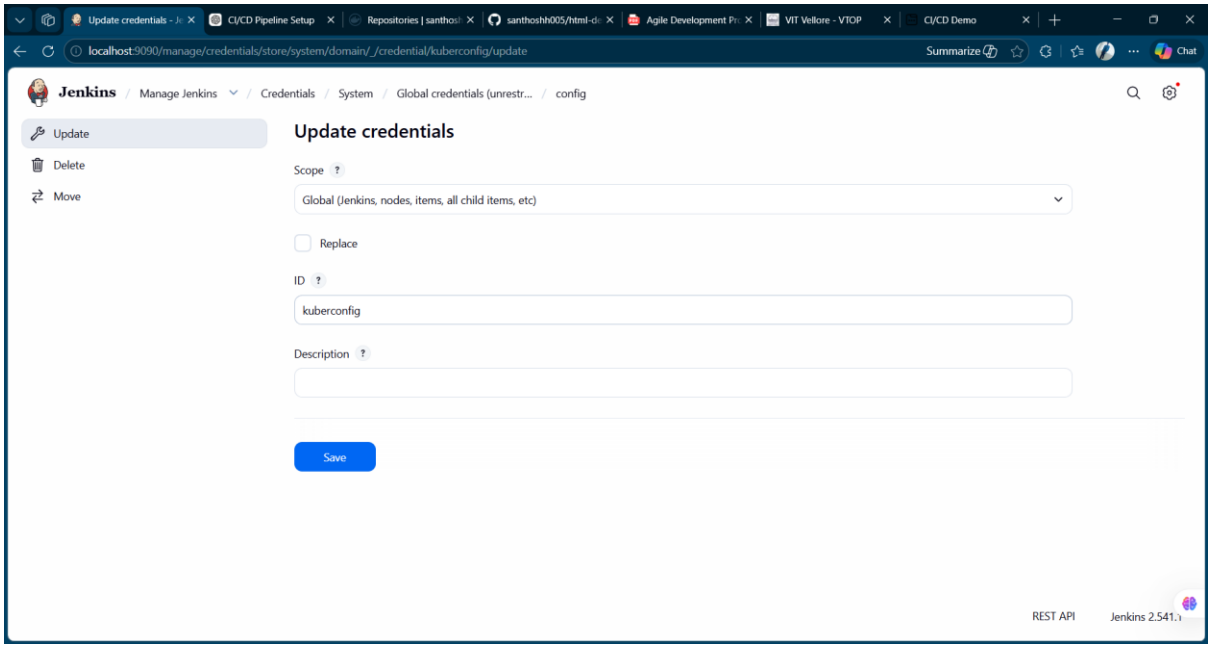
deployment.yaml:

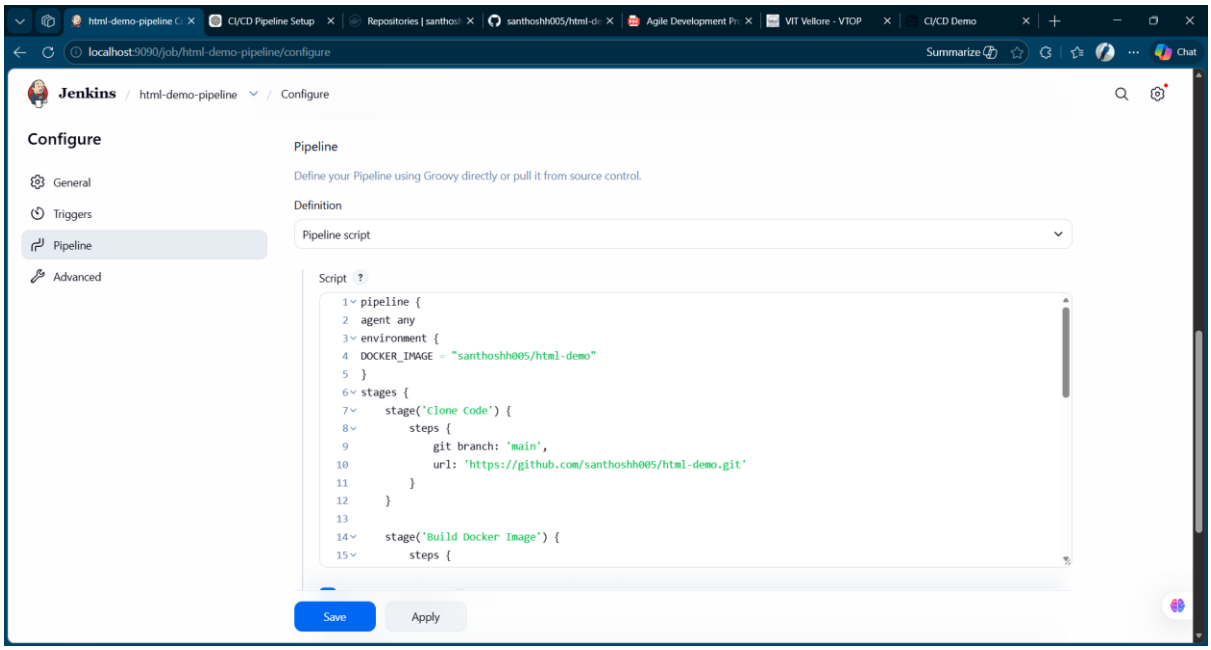
```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-demo
spec:
  replicas: 2
  selector:
```

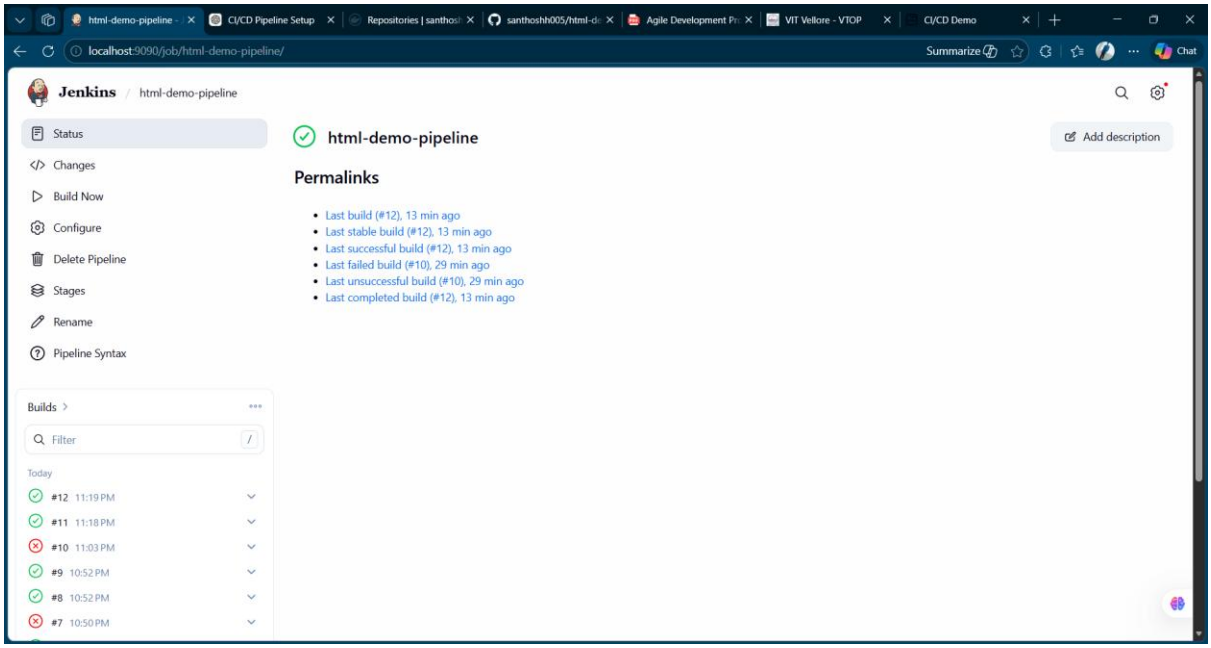
```
    matchLabels:
      app: html-demo
  template:
    metadata:
      labels:
        app: html-demo
    spec:
      containers:
      - name: html-demo
        image: santhoshh005/html-demo:latest
        ports:
        - containerPort: 80

---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
spec:
  type: NodePort
  selector:
    app: html-demo
  ports:
  - port: 80
    targetPort: 80
    nodePort: 30010
```


docker desktop PERSONAL
Search
Ctrl+K
Settings
Give feedback
Search settings
General
Resources
Docker Engine
Builders
AI
Kubernetes
Software updates
Extensions
Beta features
Notifications
Kubernetes
Enable Kubernetes
Start a Kubernetes single or multi-node cluster when starting Docker Desktop.
Cluster
docker-desktop
kind, 1 node, v1.34.3
Running
Started 37 minutes ago
Reset cluster
Cluster settings
Choose cluster provisioning method
kind
Create a cluster containing one or more nodes with kind. Requires the containerd image store
Version: 1.34.3
Changing the Kubernetes version resets your cluster. All stacks and resources are deleted.
Kubernetes version
1.34.3
Close
Apply
Engine running
Kubernetes running
RAM 2.02 GB CPU 0.44% Disk: 5.08 GB used (limit 1006.85 GB)
Terminal
v4.66.0


Update credentials - Je
CI/CD Pipeline Setup
Repositories | santhosh
santhoshh005/html-de
Agile Development Pr
VIT Vellore - VTOP
CI/CD Demo
localhost:9090/manage/credentials/store/system/domain/_/credential/dockerhub/update
Summarize
Chat
Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestr... / santhoshh005/****** (doc...
Update
Delete
Move
Update credentials
Scope
Global (Jenkins, nodes, items, all child items, etc)
Username
santhoshh005
Treat username as secret
Password
Concealed
Change Password
ID
dockerhub
Description
dockerhub
Save


Update credentials - Je
CI/CD Pipeline Setup
Repositories | santhosh
santhoshh005/html-de
Agile Development Pr
VIT Vellore - VTOP
CI/CD Demo
localhost:9090/manage/credentials/store/system/domain/_/credential/kuberconfig/update
Summarize
Chat
Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestr... / config
Update
Delete
Move
Update credentials
Scope
Global (Jenkins, nodes, items, all child items, etc)
Replace
ID
kuberconfig
Description
Save
REST API
Jenkins 2.541.1


html-demo-pipeline C
CI/CD Pipeline Setup
Repositories | santhosh
santhoshh005/html-de
Agile Development Pr
VIT Vellore - VTOP
CI/CD Demo
localhost:9090/job/html-demo-pipeline/configure
Summarize
Chat
Jenkins / html-demo-pipeline / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
pipeline {
agent any
environment {
DOCKER_IMAGE = "santhoshh005/html-demo"
}
stages {
stage('Clone Code') {
steps {
git branch: 'main',
url: 'https://github.com/santhoshh005/html-demo.git'
}
}
stage('Build Docker Image') {
steps {
Save
Apply

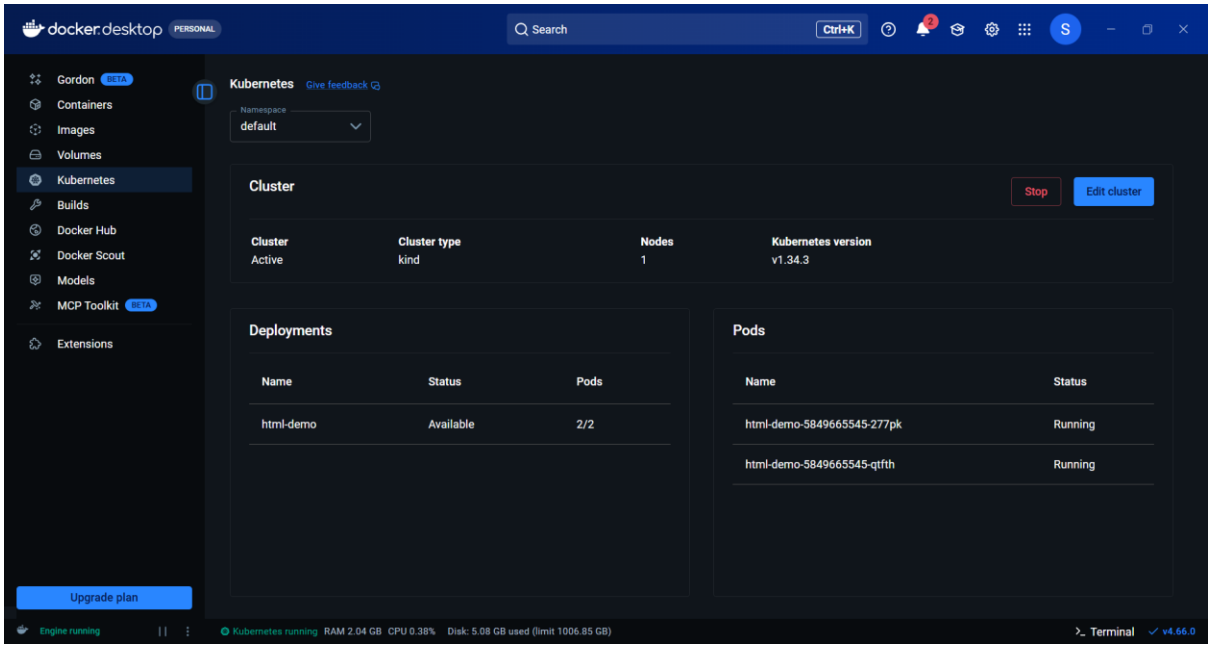


Create New Item -> Pipeline Paste the code in pipeline script - - - - Update URL of GIT repository Make sure credentialsId: “dockerhub” same as mentioned in credentials Make sure that the ID mentioned in credentials “kuberconfig” is same as in script Run the Build Pipeline:

```
pipeline {

  agent any


  environment {

    DOCKER_IMAGE = "santhoshh005/html-demo"

  }


  stages {


    stage('Clone Code') {
```

```
    steps {
        git branch: 'main',
            url: 'https://github.com/santhoshh005/html-demo.git'
    }
}

stage('Build Docker Image') {
    steps {
        bat 'docker build -t %DOCKER_IMAGE% .'
    }
}

stage('Push Image') {
    steps {
        withCredentials([usernamePassword(
            credentialsId: 'dockerhub',
            usernameVariable: 'USER',
            passwordVariable: 'PASS'
        )]) {
            bat 'docker login -u %USER% -p %PASS%'
            bat 'docker push %DOCKER_IMAGE%'
        }
    }
}

stage('Deploy to Kubernetes') {
    steps {
        withCredentials([file(
            credentialsId: 'kuberconfig',
            variable: 'KUBECONFIG'
        )]) {
```

```
            bat '''
            set KUBECONFIG=%KUBECONFIG%
            kubectl apply -f deployment.yaml --validate=false
            '''
        }
    }
  }
  }
}
```

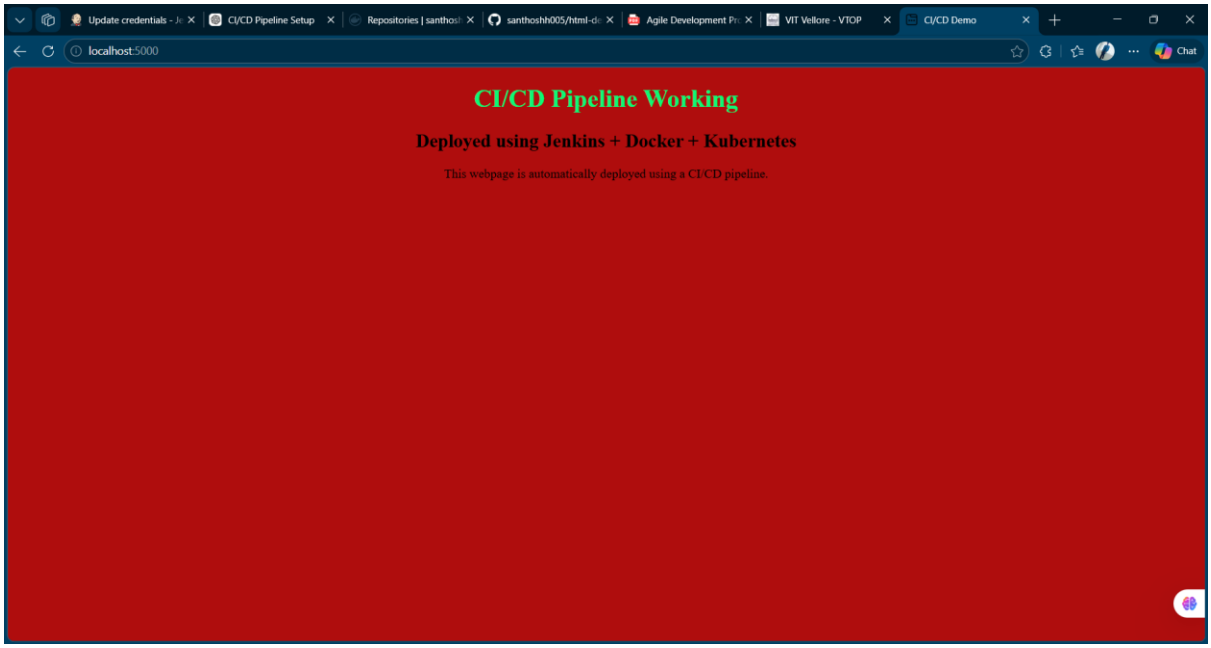


kubectl port-forward service/html-demo-service 5000:80 --address 0.0.0.0

```
PS C:\Users\saisa\.jenkins\workspace\html-demo-pipeline> kubectl port-forward service/html-demo-service 5000:80 --address 0.0.0.0
Forwarding from 0.0.0.0:5000 -> 80
Handling connection for 5000
Handling connection for 5000
Handling connection for 5000
```

localhost:5000

[CI/CD Demo]


Update credentials - Je
CI/CD Pipeline Setup
Repositories | santhosh
santhoshh005/html-de
Agile Development Pr
VIT Vellore - VTOP
CI/CD Demo
localhost:5000
Chat
CI/CD Pipeline Working
Deployed using Jenkins + Docker + Kubernetes
This webpage is automatically deployed using a CI/CD pipeline.